

## Project Progress Report:

### Implementation and Analysis of the A\* Search Algorithm

#### 1. Project Status Summary:

Our project is transitioning between the theoretical research and the experimental phases. We have managed to collect all the information necessary, making us able to develop our first working algorithm. We have now shifted our focus onto visualization and performance benchmarking.

#### 2. Completed Milestones

##### 2.1 Literature Review:

- **Scientific Base:** We are using **spaced repetition** and **memory modeling** (like FSRs) to scientifically predict when a student is likely to forget information.
- **The Problem:** Existing study algorithms are mathematically great but fail to account for **real-world constraints** like fixed class times, work, and deadlines.
- **Our Goal:** We are building the **Constrained-Path Scheduler**, using *A\* search algorithms* to find the most efficient study path that actually fits into a student's busy, non-negotiable schedule.

##### 2.2 A\* Algorithm Prototype

An experimental version of the A\* search algorithm was created, which incorporates how the algorithm expands nodes, manages a priority queue and completes creation of a full path. Within the initial set of tests run for the A\* search algorithm, results indicate that it consistently chooses the correct route over other options provided to it by users.

#### 3. Current work in progress

We have started to form a slides presentation for the final submission. We are including parts of our literary review as well as specification of our algorithm.

#### 4. Challenges and hindrances

One of our main challenges has been translating mathematical logic and low-level code into a user-friendly interface. The backend algorithm works well but visualizing the search process has been an issue. Therefore we decided to simplify the coordinate system to create a cleaner, interactive experience for the user.

#### 5. Future work

The future stages of the project will focus on testing and validation, precisely we will be:

- Testing the algorithm against randomly generated environments to evaluate reliability.
- Executing the algorithm on actual data (that we personally will collect) to determine its performance in real life environments.
- Measuring time and space complexity across different dataset sizes to determine efficiency.